

why do AI data-centres use so much water?

Posted by u/c_H_E_eeeeee_S_E • 0 points • 7 comments

Can't they just use a system like a car engine that cools down the same water and re-circulates it?

Comments

u/DiogenesKuon • 1 points • 2026-01-02 18:21 UTC

They don't. While they are often using water for cooling it uses a minuscule amount of water compared to the overall water usage of the US. The electricity usage is overstated but is a real concern, the water argument really isn't.

u/DiamondJim222 • 1 points • 2026-01-02 17:40 UTC

Car engines need much less cooling as they can withstand much higher temperatures than the electronics in a data center.

u/bangbangracer • 1 points • 2026-01-02 14:18 UTC

They do that, but it's more complicated.

What they actually use is a closed loop and an evaporative loop. The closed loop is like a car's cooling system. It recirculates a coolant mixture around the system and to a heat exchanger.

The water usage comes from the external loop. Water comes in and evaporates off the heat exchanger to carry away the heat from the building.

The internal closed loop takes heat to one place from the computers, and the external open loop takes heat from that one place through evaporation. It's like if you sprayed cold water on your car's radiator.

u/ColdAntique291 • 1 points • 2026-01-02 13:41 UTC

AI data centers generate huge amounts of heat, and water is one of the most efficient ways to remove it. Many centers do recirculate water internally, but they still lose large amounts through evaporation in cooling towers. Evaporation is what actually carries heat away, so water use stays high even in closed systems.

u/AwarenessGreat282 • 1 points • 2026-01-02 13:33 UTC

Since a data center is not moving like a car, they would then need to somehow remove the heat from the coolant before sending it back to the servers. That would take massive amounts of energy.

u/Pastadseven • 3 points • 2026-01-02 13:27 UTC

They use evaporative cooling, the water is lost to the air.

u/Strong_Soft7313 • 1 points • 2026-01-03 21:15 UTC

Yeah it's basically like how you sweat to cool down - the water has to evaporate to take the heat away, so it's gone forever unless you want to build a massive condensation system which would probably use even more energy